

MOSKVA VDNH, Russian Federation

WMO: 276120

Lat: 55.8314N

Lon: 37.6221E

Elev: 157

StdP: 99.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-18.6	-24.5	0.4	-21.6	-21.3	0.6	-17.9	4.5	-6.7	4.1	-6.1	1.1	350	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	29.9	21.1	28.0	20.2	26.2	19.4	22.1	28.4	21.1	26.5	20.1	25.1	1.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	15.0	25.2	19.0	14.1	24.1	18.2	13.3	22.9	65.4	28.5	61.7	26.6	58.2	25.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
4.1	3.4	3.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-24.5	31.6	3.4	2.4	-26.9	33.3	-28.9	34.6	-30.8	36.0	-33.3	37.7		
			-24.6	23.4	3.4	1.1	-27.1	24.1	-29.1	24.8	-31.0	25.4	-33.4	26.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-6.7	-6.3	-1.0	6.9	13.2	17.2	20.0	17.5	11.8	5.8	-0.3	-4.5	(6)
(7)		DBStd	10.53	6.17	6.39	4.55	4.98	4.89	3.75	3.46	3.66	3.88	4.22	5.07	6.00	(7)
(8)		HDD10.0	2384	519	456	341	119	24	1	0	0	24	141	310	449	(8)
(9)		HDD18.3	4580	777	689	599	343	170	66	22	59	199	388	559	708	(9)
(10)		CDD10.0	997	0	0	0	26	124	216	310	233	78	11	0	0	(10)
(11)		CDD18.3	151	0	0	0	0	12	31	73	33	2	0	0	0	(11)
(12)		CDH23.3	1222	0	0	0	3	105	255	572	274	13	0	0	0	(12)
(13)		CDH26.7	331	0	0	0	0	22	58	168	81	1	0	0	0	(13)
(14)	Wind	WSAvg	1.2	1.4	1.4	1.5	1.3	1.2	1.1	1.0	0.9	0.9	1.2	1.4	1.4	(14)
(15)	Precipitation	PrecAvg	687	46	40	35	40	54	74	86	79	65	64	54	52	(15)
(16)		PrecMax	891	98	96	88	105	150	160	192	167	183	160	146	100	(16)
(17)		PrecMin	454	3	2	4	9	0	10	4	14	12	0	4	14	(17)
(18)		PrecStd	113	21	20	18	20	32	33	44	39	32	37	23	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.2	13.5	23.5	29.2	30.5	33.5	32.9	25.1	19.1	11.6	7.8	(19)
(20)			MCWB	3.4	2.7	8.0	14.7	19.7	20.3	22.3	22.5	18.1	14.2	10.1	6.8	(20)
(21)		2%	DB	2.6	3.0	9.3	20.4	26.6	28.3	30.4	28.7	22.2	15.5	8.9	5.4	(21)
(22)			MCWB	1.8	1.7	4.9	12.9	18.3	19.9	21.6	21.0	16.8	12.6	7.8	4.4	(22)
(23)		5%	DB	1.7	2.2	6.8	17.7	23.8	26.3	28.4	26.2	19.9	13.3	7.3	3.5	(23)
(24)			MCWB	1.0	1.2	3.5	11.5	16.8	19.1	20.7	19.4	15.7	11.2	6.3	2.6	(24)
(25)		10%	DB	0.9	1.6	4.8	15.4	21.6	24.2	26.5	24.1	17.9	11.6	5.8	2.1	(25)
(26)			MCWB	0.2	0.6	2.4	10.2	15.3	18.0	20.1	18.3	14.6	10.0	4.8	1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.3	8.3	15.3	21.1	22.2	23.3	23.0	19.2	14.8	10.1	6.9	(27)
(28)			MCDWB	3.9	3.9	13.0	22.4	28.2	28.4	29.9	30.0	23.7	17.5	11.4	7.6	(28)
(29)		2%	WB	1.9	2.1	5.6	13.4	18.9	20.8	22.5	21.6	17.3	13.0	8.0	4.5	(29)
(30)			MCDWB	2.4	2.6	8.5	19.3	24.7	26.6	28.9	27.8	20.9	15.1	8.8	5.4	(30)
(31)		5%	WB	1.1	1.4	3.9	12.0	17.4	19.7	21.5	20.2	16.1	11.4	6.4	2.8	(31)
(32)			MCDWB	1.6	2.0	6.2	16.7	22.9	25.1	27.0	25.2	19.0	13.0	7.2	3.4	(32)
(33)		10%	WB	0.2	0.7	2.7	10.6	16.0	18.6	20.5	19.0	14.9	10.1	4.9	1.5	(33)
(34)			MCDWB	0.8	1.5	4.5	14.9	20.2	23.1	25.4	22.8	17.7	11.3	5.6	1.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	5.3	6.5	8.7	9.6	8.7	8.7	7.6	5.3	3.7	3.8	(35)	
(36)			MCDBR	3.5	4.2	9.3	12.6	12.7	11.9	11.7	12.0	11.2	8.1	4.8	3.5	(36)
(37)			MCWBR	3.5	3.7	5.9	6.6	5.9	5.1	4.6	5.1	6.1	5.3	4.2	3.3	(37)
(38)		5% WB	MCDBR	3.5	3.8	7.7	11.7	11.6	10.8	10.5	11.1	9.8	7.3	4.4	3.5	(38)
(39)			MCWBR	3.6	3.6	5.2	6.5	5.7	5.2	4.8	5.2	5.7	5.2	4.1	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.338	0.353	0.427	0.406	0.392	0.440	0.445	0.394	0.362	0.327	0.291	(40)	
(41)		taud	2.121	2.064	2.109	2.100	2.286	2.348	2.239	2.207	2.330	2.348	2.230	2.114	(41)	
(42)		Ebn at Noon	581	693	794	785	835	852	798	763	754	681	549	488	(42)	
(43)		Edh at Noon	61	96	117	138	121	115	127	123	95	74	56	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.58	1.44	2.97	4.00	5.26	5.47	5.38	4.34	2.76	1.26	0.48	0.31	(44)	
(45)		RadStd	0.12	0.20	0.30	0.35	0.48	0.53	0.58	0.46	0.37	0.22	0.09	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.71	N/A	N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A	(46)
(47)	Regional (10 neighbors)	+0.73	N/A	N/A	N/A	N/A	N/A	-147	-219	+110	N/A	(47)

Nomenclature: See separate page